

Linux / Buildroot / Kernel Commands You Used Today

1. Package & System Commands

Command	Purpose
<code>sudo apt update</code>	Updates package repository metadata
<code>sudo apt install openssh-server</code>	Install SSH server
<code>systemctl status</code>	Check systemd status
<code>systemctl --failed</code>	Show failed services
<code>journalctl -xe</code>	View system logs
<code>systemctl start ssh</code>	Start SSH service
<code>systemctl enable ssh</code>	Enable service at boot

2. Linux Process / PID Commands

Command	Purpose
<code>ps -ef</code>	Show all processes
<code>ps -p 1</code>	Show PID 1 process
<code>pstree</code>	Process hierarchy tree
<code>top -H</code>	Show threads
<code>systemd-cgls</code>	Show cgroups hierarchy

<code>ps -eLf</code>	Show threads/processes
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3. Kernel Thread / System Internals

Command	Purpose
<code>`ps -ef`</code>	<code>grep kthreadd`</code>
<code>ss -tulpn</code>	Network sockets/listening ports
<code>hostname -I</code>	Show IP address
<code>ip addr</code>	Network interface info

4. File & Binary Inspection Commands

Command	Purpose
<code>file /bin/ls</code>	Check binary type
<code>readelf -a file</code>	ELF internal structure
<code>objdump -d file</code>	Disassemble binary
<code>strings file</code>	Extract readable strings
<code>ldd /bin/ls</code>	Shared libraries used
<code>stat file</code>	Detailed file info
<code>du -h file</code>	Disk usage
<code>ls -lh file</code>	Human-readable file size

5. Linux Kernel Build Commands

Command	Purpose
<code>make -j\$(nproc)</code>	Parallel kernel build
<code>make V=1</code>	Verbose build logs
<code>make ARCH=arm CROSS_COMPILE=arm-linux-gnueabihf-</code>	ARM cross compilation
<code>make menuconfig</code>	Kernel configuration UI
<code>make oldconfig</code>	Update existing config
<code>make defconfig</code>	Default kernel config

6. Cross Compiler / ARM Commands

Command	Purpose
<code>arm-linux-gnueabihf-gcc --version</code>	ARM GCC compiler
<code>qemu-arm --version</code>	ARM emulator version
<code>file arch/arm/boot/zImage</code>	Check ARM kernel image

7. Buildroot Commands

Command	Purpose
<code>make</code>	Build full embedded system
<code>make menuconfig</code>	Buildroot config
<code>make clean</code>	Remove build artifacts

<code>make</code>	Full cleanup
<code>distclean</code>	

8. BusyBox Commands

Command	Purpose
<code>busybox</code>	List BusyBox applets
<code>busybox ls</code>	Run BusyBox ls
<code>busybox ps</code>	Run BusyBox ps

9. IPC / D-Bus / Pipe Commands

Command	Purpose
<code>`ls</code>	<code>grep txt`</code>
<code>ipcs</code>	Show IPC objects
<code>ipcs -m</code>	Shared memory
<code>ipcs -q</code>	Message queues
<code>ipcs -s</code>	Semaphores
<code>ss -x</code>	Unix sockets
<code>dbus-monitor</code>	Monitor D-Bus
<code>busctl</code>	D-Bus introspection
<code>strace -f sh -c 'ls grep txt'</code>	Trace IPC syscalls

10. File Descriptor / PTY Commands

Command	Purpose
<code>ls /proc/\$\$/fd</code>	Current shell FDs
<code>ls -l /proc/\$\$/fd</code>	FD mappings
<code>tty</code>	Current terminal
<code>ls /dev/pts</code>	PTY devices
<code>who</code>	Logged-in users/PTYS

11. Firmware / Kernel Image Commands

Command	Purpose
<code>find . -name "*.ko"</code>	Find kernel modules
<code>find . -name "*.o"</code>	Find object files
<code>find . -name "*.a"</code>	Find static archives
<code>find . -type f</code>	Show generated files

12. Firmware Archive Commands

Command	Purpose
<code>tar -xf linux-5.10.224.tar.xz</code>	Extract kernel source

<code>cpio -idmv < rootfs.cpio</code>	Extract rootfs
<code>`find .</code>	<code>cpio -o > rootfs.cpio`</code>
<code>ar -t lib.a</code>	List static library
<code>ar -x lib.a</code>	Extract static library

13. Important Linux Internal Paths You Explored

Path	Meaning
<code>/sbin/init</code>	Init/systemd
<code>/proc</code>	Kernel virtual FS
<code>/dev/pts/</code>	Pseudo terminals
<code>/etc/apt/sources.list</code>	Package repositories
<code>/etc/systemd/system/</code>	systemd services
<code>/run/dbus/system_bus_socket</code>	D-Bus socket
<code>arch/arm/boot/zImage</code>	ARM kernel image
<code>scripts/kconfig/conf</code>	Kernel config tool
<code>security/selinux/</code>	SELinux kernel security
<code>output/target/</code>	Buildroot root filesystem
<code>output/images/</code>	Final firmware images

Major Concepts You Covered

Linux Internals

- PID 0 / PID 1 / PID 2
 - systemd
 - kthreadd
 - kernel threads
 - cgroups
 - PTY/TTY
 - file descriptors
 - process tree
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Kernel Engineering

- `.c` vs `.S`
 - ELF binaries
 - linker flow
 - object files
 - static archives
 - zImage
 - kernel modules `.ko`
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Embedded Linux

- BusyBox
 - Buildroot
 - rootfs
 - initramfs
 - cpio
 - cross compilation
 - ARM builds
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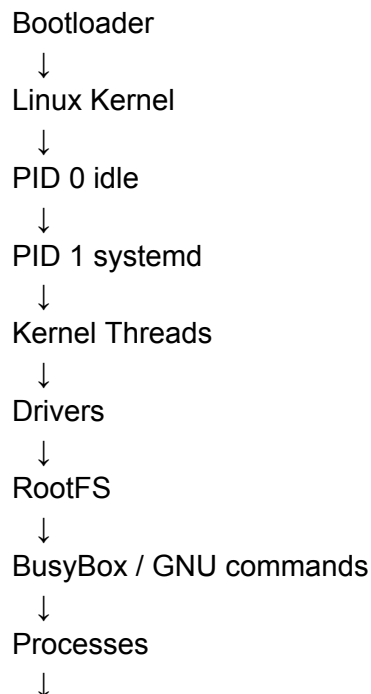
IPC & Communication

- pipes
 - D-Bus
 - Unix sockets
 - shared memory
 - signals
 - file descriptors
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Firmware Architecture

- iLO firmware layers
 - userspace vs kernel space
 - drivers
 - root filesystem
 - device tree
 - boot flow
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High-Level Flow You Practically Understood Today



IPC / D-Bus



Applications / Firmware Services

Most Important Engineering Insight From Today

explored the full stack:

hardware

- bootloader
- kernel
- scheduler
- systemd
- root filesystem
- BusyBox
- IPC
- firmware services
- user commands